Sorry brooo 😭 The text block clearly isn't displaying on your side. I'll put the full SurgiGuard AI project prompt directly here as plain text, so you can see and copy it.

You can paste this into ChatGPT, Claude, Codex, or any other AI development tool.

# MASTER PROMPT — SURGIGUARD AI

## 1. Project Overview

Project Name: SurgiGuard AI Domain: Healthcare, Artificial Intelligence, IoT, Digital Health Context: HackCure Healthcare Hackathon Tagline: Smarter Recovery. Safer Care.

### Project Vision

SurgiGuard AI is an intelligent postoperative recovery platform that accompanies patients throughout their recovery after surgery.

It connects three essential participants:

- Patients, through a personalized mobile application.
- Trusted caregivers, such as family members or friends, through a separate account in the same mobile application.
- Medical professionals, through a dedicated web dashboard.

The system combines affordable IoT health monitoring, artificial intelligence, personalized recovery plans, medication management, rehabilitation support, and clinical follow-up coordination.

Its main purpose is not simply to detect abnormal vital signs, but to understand whether a patient is recovering as expected, investigate concerning changes, and help ensure that appropriate medical follow-up takes place.

## 2. The Problem

Following surgery, patients often return home with prescriptions, discharge instructions, rehabilitation recommendations, and scheduled medical appointments.

However, postoperative recovery presents several challenges:

- Complications may develop between medical appointments.
- Patients may not recognize warning signs early enough.
- Medical teams have limited visibility into recovery at home.
- Patients may forget medications or misunderstand recovery instructions.
- Family members may struggle to provide appropriate support.

- Traditional monitoring approaches may rely on isolated measurements rather than personalized recovery progress.
- Even when alerts are generated, appropriate follow-up may not occur.

SurgiGuard aims to bridge the gap between hospital discharge and safe, coordinated recovery at home.

# 3. Our Solution

SurgiGuard is a connected recovery ecosystem supported by intelligent clinical decision assistance.

Its core workflow is:

Monitor → Detect → Investigate → Escalate → Act → Verify → Continue Recovery

The system evaluates physiological measurements, patient-reported symptoms, medical history, surgical information, prescribed medications, and clinician-approved recovery plans.

When a concerning change occurs, SurgiGuard investigates the available evidence, explains potential concerns and uncertainty, and follows appropriate clinical escalation protocols.

Medical decisions remain under the supervision of qualified professionals.

# 4. Platform Structure

## A. Patient Space — Mobile Application

Each patient has a personalized recovery environment.

Recovery Dashboard

- Personalized recovery timeline.
- Current postoperative recovery stage.
- Daily recovery objectives.
- Recovery progress and milestones.
- Expected versus observed recovery trajectory.
- Upcoming medical appointments.

IoT Health Monitoring

- Connected heart-rate monitoring.
- Temperature measurements.
- Activity and mobility monitoring where supported.
- Historical trends.
- Notifications about clinically relevant changes.

Daily Symptom Check-ins

- Pain level.
- Fatigue.
- Wound-related symptoms.
- Mobility difficulties.
- Other symptoms relevant to the surgical procedure.

AI Recovery Assistant

- Answers recovery-related questions using verified medical information.
- Explains discharge instructions.
- Helps patients understand their medications.
- Provides guidance based on their clinician-approved recovery plan.
- Recognizes when professional assessment is necessary.

Medication Safety and Adherence

- Personalized medication schedules.
- Medication reminders.
- Dose confirmation.
- Missed-dose tracking.
- Potential interaction, duplication, and allergy-conflict screening using trusted references.
- Clinician-verified medication guidance.

Rehabilitation

- Personalized, approved rehabilitation exercises.
- Activity reminders.
- Daily objectives.
- Progress tracking.
- Recovery milestones.

Communication

- Assistance requests.
- Caregiver communication.
- Recovery notifications.
- Relevant clinical follow-up updates.

## B. Caregiver Space — Mobile Application

A trusted family member, friend, or caregiver can create a separate account and connect with the patient through an invitation.

The patient controls which information is shared.

Caregiver Dashboard

- Authorized recovery updates.
- Upcoming appointments.

- Medication reminders where permitted.
- Rehabilitation tasks.
- Patient assistance requests.

Caregiver Notifications

- Missed recovery check-ins.
- Assistance requests.
- Relevant recovery concerns.
- Important follow-up reminders.

Caregiver Support

- Help patients complete everyday recovery activities.
- Confirm practical assistance.
- Report observed changes or difficulties.
- Support medication and appointment adherence.

Caregivers do not make medical decisions. The system must remain functional even if a patient has no caregiver.

## C. Medical Team Space — Web Dashboard

Doctors, nurses, and other authorized healthcare professionals access SurgiGuard through a dedicated web dashboard.

Patient Management

- View assigned postoperative patients.
- Access relevant patient records.
- Review surgery information.
- Monitor recovery status.
- Prioritize patients requiring attention.

Clinical Monitoring

- View IoT measurements.
- Analyze symptoms and recovery trends.
- Compare expected and observed recovery.
- Identify potentially concerning deviations.

AI Clinical Investigation

- Review evidence-grounded alerts.
- Understand why an anomaly was detected.
- Inspect sensor reliability.
- Review uncertainty and conflicting evidence.
- Assess patient-specific context.

Recovery Plan Management

- Upload surgical and discharge reports.
- Review AI-generated recovery protocols.
- Approve or modify proposed instructions.
- Manage medication schedules and rehabilitation plans.

Clinical Follow-up

- Acknowledge alerts.
- Record medical decisions.
- Assign follow-up tasks.
- Communicate updated instructions.
- Verify follow-up completion.

The dashboard supports simultaneous monitoring and management of multiple patients.

# 5. Core AI Innovations

## Innovation 1 — Patient Recovery Twin

SurgiGuard creates a personalized computational representation of each patient's expected recovery.

The Recovery Twin considers:

- Type of surgery.
- Time since surgery.
- Relevant medical information.
- Clinician-approved recovery milestones.
- Physiological measurements.
- Symptoms.
- Mobility and rehabilitation progress.
- Medication adherence.

It compares expected recovery with actual recovery over time.

Rather than relying only on fixed thresholds, the system identifies meaningful changes in recovery patterns.

For the MVP, this can be implemented using approved recovery pathways, transparent statistical models, and simulated patient scenarios.

## Innovation 2 — Uncertainty-Aware AI Investigation

When an anomaly appears, SurgiGuard investigates the available information.

It examines:

1. Sensor reliability and measurement quality.
2. Patient symptoms.

3. Historical measurements.
4. Expected recovery progress.
5. Relevant clinical context.
6. Missing or contradictory information.
7. Approved warning signs and escalation criteria.

The system produces an explainable investigation summary.

It must not independently diagnose complications or delay urgent clinical escalation while gathering additional information.

## Innovation 3 — Recovery Protocol Compiler

Medical professionals upload surgical reports, prescriptions, and discharge documents.

AI extracts relevant information and proposes a structured recovery plan containing:

- Monitoring schedules.
- Medication reminders.
- Symptom check-ins.
- Rehabilitation exercises.
- Recovery milestones.
- Follow-up appointments.
- Warning signs.

The medical team reviews and approves the proposed plan before it becomes active.

## Innovation 4 — Closed-Loop Recovery Verification

SurgiGuard does not stop after generating an alert.

It tracks:

Abnormal Observation → Clinical Alert → Acknowledgment → Medical Decision → Updated Instructions → Follow-up Verification

The goal is to prevent important care tasks from being overlooked.

The system distinguishes between sending an alert, acknowledging it, and completing the required follow-up.

## Innovation 5 — Collaborative Recovery

SurgiGuard connects patients, caregivers, and medical professionals through one coordinated recovery workflow.

Patients complete check-ins and follow recovery instructions.

Caregivers offer practical assistance and share authorized observations.

Medical professionals supervise clinical decisions.

AI helps organize information, identify concerning changes, and coordinate the approved workflow.

# 6. IoT Integration

The HackCure MVP should integrate affordable hardware.

Possible components:

- ESP32 microcontroller.
- Compatible temperature sensor.
- Heart-rate sensor.
- Optional movement or activity sensor.

IoT measurements are transmitted to the backend and visualized in the patient application and medical dashboard.

Hobby-grade sensors are used for demonstrating technical integration, not for validated medical diagnosis.

# 7. Example Patient Journey

Day 0 — Hospital Discharge

A patient undergoes surgery.

The medical team uploads their discharge documentation. SurgiGuard generates a proposed personalized recovery protocol.

The doctor reviews and approves it.

Day 1 — Recovery at Home

The patient opens the SurgiGuard mobile application.

They review medication reminders, rehabilitation instructions, and daily recovery objectives.

Their caregiver can access authorized information and assistance tasks.

Day 3 — Recovery Monitoring

SurgiGuard collects available sensor measurements and symptom reports.

The Recovery Twin compares observed recovery with expected milestones.

Day 5 — Concerning Change

The patient reports worsening pain and fatigue. An elevated temperature is also recorded.

SurgiGuard examines measurement reliability, symptoms, recent trends, and the approved recovery protocol.

If escalation criteria are met, the system alerts the medical team.

Clinical Intervention

The doctor reviews the alert and evidence through the medical dashboard.

The doctor contacts the patient and records appropriate instructions.

Follow-up Verification

SurgiGuard tracks whether the patient received the instructions and whether the necessary follow-up was completed.

Recovery monitoring continues.

# 8. Suggested Technical Architecture

Mobile App: Flutter or React Native.

Medical Dashboard: React or Next.js.

Backend: Python and FastAPI.

Database: PostgreSQL.

Artificial Intelligence:

- LLM-based medical document extraction.
- Retrieval-augmented generation using verified references.
- Recovery trajectory analysis.
- Signal reliability evaluation.
- Controlled AI investigation workflows.
- Structured clinical rules for escalation.

IoT: ESP32 with suitable low-cost sensors.

Security: Authentication, role-based access control, caregiver permissions, encrypted communications, and audit logs.

# 9. HackCure MVP

The working prototype should demonstrate:

1. A patient mobile interface with recovery monitoring.

2. A separate caregiver interface with authorized access.
3. A medical web dashboard.
4. Live IoT sensor data transmission.
5. Personalized recovery-plan generation from a sample medical document.
6. Clinician review and approval.
7. Expected-versus-observed recovery visualization.
8. AI investigation of a simulated postoperative complication.
9. Medical alert notification and acknowledgment.
10. Verified follow-up completion.

The demonstration should use fictional patient records and explicitly label simulated medical measurements.

# 10. What Makes SurgiGuard Innovative?

SurgiGuard does not claim to invent patient monitoring, medication reminders, or AI healthcare assistance.

Its proposed differentiation comes from combining:

- Personalized recovery trajectory modeling.
- Evidence-grounded clinical investigation.
- Explicit uncertainty handling.
- Clinician-approved recovery protocols.
- Patient-caregiver-clinician collaboration.
- Closed-loop medical follow-up.
- Affordable IoT integration.

The central concept is Evidence-to-Action Recovery Intelligence.

Instead of simply collecting measurements and generating alerts, SurgiGuard seeks to connect observations to meaningful, accountable care.

# 11. Long-Term Vision

SurgiGuard aims to evolve into a platform supporting multiple types of surgery and increasingly personalized postoperative recovery.

Potential future features include validated prediction models, integration with hospital electronic health records, advanced rehabilitation analytics, multilingual patient support, and assistive communication tools such as SilentVoice for patients with temporary speech difficulties.

Clinical safety, privacy, accessibility, and medical oversight remain essential throughout development.

# 12. Final Pitch

SurgiGuard AI — Smarter Recovery. Safer Care.

SurgiGuard AI is an intelligent postoperative recovery and care coordination platform connecting patients, trusted caregivers, and medical professionals.

By combining affordable IoT monitoring, personalized recovery modeling, evidence-grounded AI investigation, and closed-loop follow-up, it aims to make postoperative recovery safer, more personalized, and better coordinated.

We don't just monitor recovery. We help transform recovery information into coordinated care.

That's the complete prompt, directly in the chat this time—no special document block required.